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NeuroAgent

**A clinical collaboration proposal:
validating an evidence-driven neurological diagnosis benchmark**

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8 July 2026

A new way of testing medical AI

Most research today tests medical AI with **question answering**. Real clinical work is not like that.

Today: question answering

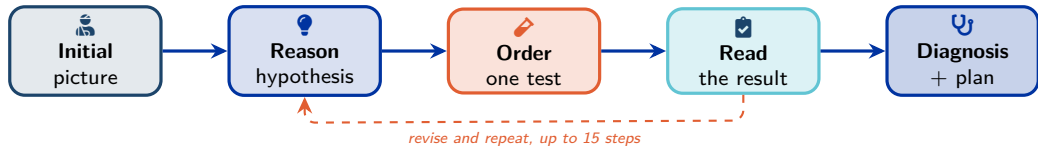
- Multiple-choice exams
- Every clue is already in the question
- One correct option to pick
- Rewards recall, not reasoning

Real clinical practice

- The patient arrives with a complaint, not options
- You decide which investigations to order
- Results come back over time
- Cost, risk and safety all matter

We introduce a **new paradigm**: test the AI on the **whole diagnostic workup**, the way a clinician actually reasons.

Project overview



600 realistic neurology cases across 20 conditions.

What we measure

- Right test, right order, sensible cost
- Sound reasoning, not memorised facts
- A workup a clinician can follow

Why it is new

- Today's AI exams hand over every clue
- Tests the real skill: what to investigate
- Each case checked by **2+ clinicians**

The agent in action

The goal: a transparent workup, ending in a diagnosis backed by evidence.

The screenshot displays the NeuroAgent interface, which is divided into several panels. On the left, a sidebar contains navigation options: Cases, Dataset, Traces, Rules, and Settings. Below these is a search bar and a list of cases, including 'ALS-H01', 'ALS-H02', 'ALZ-EARLY-H02', 'ALZ-EARLY-H03', 'ALZ-EARLY-H01', and 'FXT1-T00-H02'. The main panel shows a 'Trace Replay' for 'ALZ-EARLY-H01'. It includes patient information: '68yo male', 'right-handed', 'Hispanic', 'BMD 273', 'outpatient', and '97%'. Below this is a 'CHIEF COMPLAINT' section: 'Subjective memory difficulties and word-finding trouble over the past year'. The 'HISTORY OF PRESENT ILLNESS' section follows, detailing the patient's symptoms and medical history. On the right, the 'Agent Timeline' shows a sequence of reasoning steps: 'Agent Reasoning' (1), 'Lab Results' (2), 'Agent Reasoning' (3), 'Brain MRI' (4), 'Agent Reasoning' (5), 'Drug Interactions' (6), and 'Final Assessment' (7). The 'Final Assessment' section provides a 'Primary Diagnosis' of 'Mixed Dementia (Alzheimer Disease + Vascular Cognitive Impairment)' with a confidence of 0.80, and lists 'Differential Diagnoses' and 'Key Evidence'.

The dataset

Cases

600

20 conditions

Per condition

30

exactly, fully balanced

Follow-up results

3,800

triggered by actions

How each case is built

- Patient core: history, exam, vitals
- Evidence hidden behind tool calls
- Ground truth: best path, red herrings, unsafe actions

Spanning

- 3 difficulty tiers: straightforward to puzzle
- 396 synthetic + 204 real-report-seeded
- 3 settings: emergency / inpatient / outpatient

*Results are cached in two tiers: **initial** findings when a test is first ordered, and **follow-ups** unlocked by a specific next action.*

How the cases were generated

Two pipelines, one schema: real cases for fidelity, synthetic cases for scale.

Seeded from real case reports

- Source: **MedCaseReasoning**, clinician-written case reports
- Rewritten by **Claude Opus 4.8 + Sonnet 4.6**: findings hidden in tool outputs

204 cases

Fully synthetic, for scale

- Built from per-condition clinical schemas, no source report
- Scale needed to **fine-tune local models**
- Same schema and ground truth as the seeded cases

396 cases

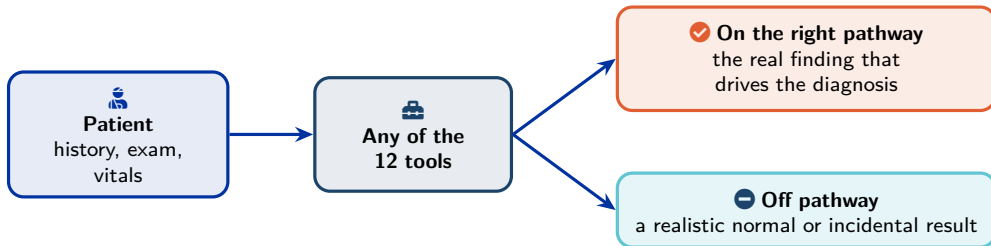
The 12 diagnostic tools

Tool	What it returns
Brain MRI	report with region-tagged findings
EEG	background, epileptiform features, focality
ECG	rhythm, intervals, signs of ischaemia
Labs	panels with reference ranges + abnormality flags
CSF (lumbar puncture)	cells, protein, glucose, NfL, oligoclonal bands
CT scan	non-contrast / contrast, head / spine
Echocardiogram	chambers, valves, ejection fraction, thrombus
Cardiac monitoring	telemetry / Holter / loop recorder
Advanced imaging	MRA / MRV / CTA / PET / DaT-SPECT
Specialised tests	EMG/NCS, biopsy, autoimmune panels
Literature search	guideline and paper snippets
Drug interactions	interaction profile + contraindications

Each returns a structured, typed report with a cost; any tool, any time.

Every patient is a complete picture

Each case pre-computes a **whole patient**, so the agent can order **any** of the 12 tools and always get a realistic answer.



Never a dead end: every tool answers. But each call costs money and a turn, so ordering the wrong test is **penalised**.

A worked example: wake-up ischemic stroke

Turn 1: what the agent sees

- 58 F: woke with slurred speech, clumsy hand
- Last known well 9 h ago: window unclear
- Left face/arm weak, sensory loss, neglect

What the right workup reveals

- CT first: rule out bleed; glucose: no hypoglycemia
- MRI: fresh infarct, deep right white matter
- ECG, echo, carotids, Holter: hunt the cause

The trap, and an off-pathway test

“Chronic, not acute” tempts an early stop, yet the cause hunt is still owed. An off-pathway EEG reads **normal**: wasted.

📌 **Ground truth:** acute ischemic stroke, right corona radiata (ICD-10 I63.541).

How we evaluate performance

Automatic metrics, per case

- **Diagnosis:** ICD top-1 / top-3 match
- **Tool selection:** precision, recall, F1 + coverage
- **Safety:** contraindicated, harmful, wrong-order calls
- **Cost:** \$ vs optimal; useless or repeated calls

AI reasoning judge, 0–5 each

- Diagnostic accuracy; evidence found + integrated
- Differential reasoning; uncertainty calibration
- Tool efficiency, safety, red-herring handling
- One **composite reasoning score**

Clinician validation: the gold standard

≥2 clinicians per case: field-level approval + inter-rater agreement, the human ceiling we calibrate against.

Goals for the medical collaboration

1. Validate the cases

- Read each patient as you would in clinic
- Approve, flag, or correct field by field
- Drop cases that are not good enough

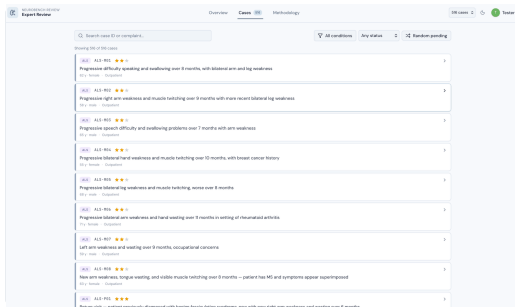
2. Validate our scoring

- Score a sample of agent runs yourself
- Compare against the AI reasoning judge
- Confirm scores track clinical quality

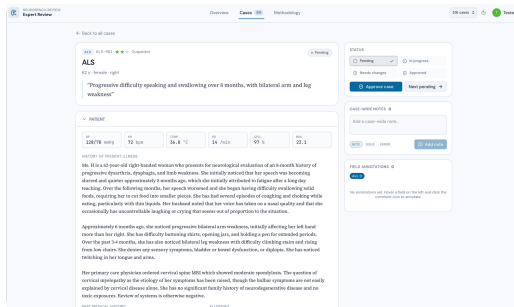
How the review works

- **≥ 2 independent reviewers** per case
- Separate login per reviewer: blind, no shared bias
- Agreement stats → fix or drop weak cases → higher quality

The annotation platform



 Browse cases: condition, difficulty, status



 One screen per patient: vitals, history, full workup

 review.andreaprotani.com

Reviewing a case

PAST MEDICAL HISTORY

Hypertension, diagnosed 10 years ago, well-controlled on lisinopril

Hypothyroidism, on levothyroxine for 8 years

Osteoporosis, on calcium and vitamin D supplementation

PATIENT > CLINICAL HISTORY

> PAST MEDICAL HISTORY > [2]

Note

Issue

Error

Describe the issue or note...

⌘+ to save - Esc to dismiss

Cancel

Add

Alcohol Social

1-2 glasses of wine per week

Living

Lives with husband, two adult children

Situation

nearby

Functional

Previously fully independent; now difficulty

ALLERGIES

Penicillin — rash

MEDICATIONS

Lisinopril · 10 mg · once daily
for Hypertension

Levothyroxine · 75 mcg · once daily in the morning
for Hypothyroidism

Calcium carbonate · 600 mg · twice daily with meals
for Osteoporosis prevention

Cholecalciferol · 1000 IU · once daily
for Vitamin D supplementation

STATUS

Pending

In progress ✓

Needs changes

Approved

Approve case

Next pending →

CASE-WIDE NOTES 0

Add a case-wide note...

NOTE

ISSUE

ERROR

Add note

FIELD ANNOTATIONS 1

ALL 1

ISSUE 1

patient.history_present_illness

age not coherent with the pathology

ISSUE

🗨 Flag any field: note, issue, or error

☰ Per-case status, every annotation
logged

Approve, request changes, or move on: progress tracked per reviewer.



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